Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 10: For All, There Exists

Quantified Statements

Chartrand & Zhang, Section 3.1, pp. 79–84

Name: _____ Date: _____

A loose end from Day 1, and a new unit

On the very first day we met open sentences — "$x + 2 = 7$" — and said they are not statements, because they have no truth value until the variable is fixed. Choosing a value is one way to fix that. Today we meet the other: saying something about *all* values, or about *some* value.
Logic is behind us. This is where the rest of the course starts.

By the end of today you should be able to

1. Write quantified statements using $\forall$ and $\exists$.
2. Decide whether a quantified statement is true or false, and know what that costs.
3. Negate a quantified statement, including one with two quantifiers.

1. The universal quantifier

Universal quantifier

Let $R(x)$ be an open sentence over a domain S. The phrases *for all*, *for every*, *for each* are the **universal quantifier**, written $\forall$. The sentence

$$\forall x \in S,\ R(x)$$

is a **statement**, read "for every x in S, $R(x)$."

Example 3.1. For $R(n)$: $n^2 - n$ is even, over $\mathbb{Z}$:

$$\forall n \in \mathbb{Z},\ R(n): \quad \text{for every integer } n,\ n^2 - n \text{ is even}$$

A universal statement is an implication in disguise

$\forall x \in S,\ R(x)$ can always be rephrased as

$$\text{If } x \in S, \text{ then } R(x).$$

So "for every integer n, $n^2 - n$ is even" and "if n is an integer, then $n^2 - n$ is even" say the same

thing. Everything you learned about implications in Week 2 carries over unchanged.

2. The existential quantifier

Existential quantifier

The phrases *there exists*, *there is*, *for some*, *for at least one* are the **existential quantifier**, written $\exists$. For an open sentence $Q(x)$ over S,

$$\exists x \in S,\ Q(x)$$

is read "there exists x in S such that $Q(x)$."

Statement	T/F	Why
$\forall x \in \mathbb{R},\ x^2 > 0$	F	Fails at $x = 0$.
$\exists n \in \mathbb{Z},\ n^2 = n$	T	$n = 0$ works (so does $n = 1$).
$\exists x \in \mathbb{R},\ x^4 + 2 = 2x^2$	F	$u^2 - 2u + 2 = 0$ has no real root.

The two jobs cost very different amounts

	To show TRUE	To show FALSE
$\forall x \in S, R(x)$	argue for *every* x	one counterexample
$\exists x \in S, R(x)$	one example	argue for *every* x

Decide which box you are in *before* you start. It tells you whether you owe an example or an argument.
You met this asymmetry in Week 1 with $n^2 - n + 41$, and again in Week 3 with satisfiability. This is the same idea a third time, now with symbols for it.

3. Negating a quantified statement

The two negation rules

$$\sim\big(\forall x \in S,\ R(x)\big) \ \equiv\ \exists x \in S,\ \sim R(x)$$
$$\sim\big(\exists x \in S,\ R(x)\big) \ \equiv\ \forall x \in S,\ \sim R(x)$$

The quantifier flips and the open sentence is negated.

This is De Morgan, one more time

Think of $\forall$ as a very long **conjunction** — $R(a_1) \wedge R(a_2) \wedge \cdots$ — and $\exists$ as a very long **disjunction**. Then these rules are exactly $\sim(P \wedge Q) \equiv (\sim P) \vee (\sim Q)$ and its partner, stretched over a whole domain. Practice Set 3 Problem 12 promised this would come back. Here it is.

Negate each. Say it the way a person would.

Statement	Negation
Everyone likes "The Wizard of Oz."	There is someone who does not like it.
$\forall n \in \mathbb{Z}$, $2n^2 + 5n + 1$ is odd.	There is an integer n with $2n^2 + 5n + 1$ even.
$\exists x \in \mathbb{R}$, $x^2 < 0$.	For every real x, $x^2 \geq 0$.

The overshoot — the most common error in this section

The negation of "everyone likes it" is **not** "no one likes it." It is **"at least one person does not like it."**
"No one likes it" is $\forall x, \sim R(x)$ — a far stronger claim, and the negation of a *different* statement. Overshooting turns an easy disproof into an impossible burden.

4. Two quantifiers at once

Order matters. Read left to right and watch what depends on what.

Statement	T/F	Why
$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z},\ x + y = 0$	T	Given x, take $y = -x$.
$\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z},\ x + y = 0$	F	No single y works for every x.

Same symbols, same open sentence, opposite truth values. In the first, y may depend on x; in the second it must be chosen in advance.

Negating a mixed statement

Flip every quantifier, left to right, and negate the open sentence at the end:

$$\sim\big(\forall x \in S, \exists y \in T,\ R(x,y)\big) \ \equiv\ \exists x \in S, \forall y \in T,\ \sim R(x,y)$$

Exit ticket

1. Write the negation of "Every student in CS 341 has taken CS 204."

2. Is $\exists n \in \mathbb{Z},\ n^2 = 2$ true or false? Justify in one line.

3. What must you produce to disprove a statement beginning $\forall$?

Before Friday

Read Chartrand Section 2.1, "Sets and Subsets," pp. 55–60, and Section 2.3, "Cartesian Products," pp. 70–72.

Self-check (not collected). Section 3.1, Exercises 1, 3, 5, 7, and 13 — all odd-numbered, so answers are in the back. Attempt them fully first.

Coming up

Homework 4 is due Friday, September 11, 11:59 p.m. — Section 1.5 and normal forms.
Quiz 4 is Monday, September 14, on Sections 3.1 and 2.1.

Questions I still have